

Part VI Homework — D.9(a), D.10, F.9, F.10

Concise summary with clean math, run commands, and result placeholders.

Run Commands (what to execute)

- Mass PD (D.8, for reference):
 - `python _D_mass/python/hwD8_massSim.py`
- Mass digital PID (D.10):
 - `python _D_mass/python/hwD10_massSim.py`
- VTOL nested PID (F.10):
 - Controller: `hw1_s/F_planar_vtol/ctrlNestedPID_digital.py:1`
 - Ask to add a small VTOL sim runner if needed.

Paste Your Output Images Here

- D.9(a) type/SSE: [Paste image]
- D.10 mass PID step/disturbance: [Paste image]
- F.9 VTOL type summaries: [Paste image]
- F.10 nested PID step in h and z: [Paste image]

D.9(a) — System Type & Integrators (Mass–Spring–Damper)

Unity feedback error transfer and final value theorem:

$$E(s) = \frac{1}{1 + P(s)C(s)} R(s), \quad e_{\infty} = \lim_{s \rightarrow 0} sE(s).$$

- PD: $C_{PD}(s) = k_P + k_D s \rightarrow$ type 0 (no free integrators).
 - Step: finite SSE; Ramp: ∞ ; Parabola: ∞ .
- Add integrator: $C_{PID}(s) = k_P + \frac{k_I}{s} + k_D s \rightarrow$ type 1.
 - Step: 0 SSE; Ramp: finite; Parabola: ∞ .

Takeaway: a single integrator guarantees zero SSE to steps (more robust to plant uncertainty).

D.10 — Digital PID (Mass–Spring–Damper)

Continuous-time form with dirty derivative ($\sigma = 0.05$):

$$C(s) = k_P + k_I \frac{1}{s} + k_D \frac{s}{\sigma s + 1}.$$

Position form: $u = k_P e + I + k_D \dot{e}_f$, with $e = r - y$.

Integral discretizations (sample time T_s):

- Backward-Euler:

$$I[k] = I[k-1] + k_I T_s e[k].$$

- Tustin (trapezoidal):

$$I[k] = I[k-1] + \frac{k_I T_s}{2} (e[k] + e[k-1]).$$

Filtered derivative (dirty derivative):

$$\dot{e}_f[k] = \frac{\dot{e}_f[k-1] + \frac{e[k] - e[k-1]}{T_s}}{1 + \frac{\sigma}{T_s}}.$$

Measurement-only difference equations (position form):

```
e[k]   = r[k] - y[k]
I[k]   = I[k-1] + k_I*T_s*e[k]           # or trapezoidal form above
dhat[k]= (dhat[k-1] + (e[k]-e[k-1])/T_s) / (1 + sigma/T_s)
u[k]   = k_P*e[k] + I[k] + k_D*dhat[k]
```

Files:

- Controller: `_D_mass/python/ctrlPID_digital.py:1`
- Simulation: `_D_mass/python/hwD10_massSim.py:1`

Checklist:

- Step tracking: SSE ≈ 0 with $k_I > 0$.
- Parameter uncertainty ($\pm 20\%$): PID removes bias that PD leaves.
- Reasonable overshoot/settling; derivative filter tames noise.

F.9 — System Type & Integrators (VTOL)

- Altitude (h), PD $\rightarrow$ type 0; add I in altitude loop $\rightarrow$ type 1 (step 0 SSE).
- Attitude (θ) inner loop, PD $\rightarrow$ type 0 w.r.t. θ command; constant torque disturbance $\rightarrow$ finite SSE; usually no I here.
- Lateral position (z) outer loop, PD $\rightarrow$ type 0; add I in outer loop $\rightarrow$ type 1 (step 0 SSE).

F.10 — Digital Nested PID (VTOL)

Architecture:

- Inner attitude (θ): PD with dirty derivative (fast loop).
- Outer altitude (h): PID with integrator.
- Outer lateral (z): PID producing θ_{cmd} with integrator.

Discrete-time skeleton:

```
# altitude (h)
e_h[k]  = h_r[k] - h[k]
I_h[k]  = I_h[k-1] + kI_h*T_s*e_h[k]
dhat_h[k]= (dhat_h[k-1] + (e_h[k]-e_h[k-1])/T_s)/(1+sigma/T_s)
F_dev[k] = kP_h*e_h[k] + I_h[k] + kD_h*dhat_h[k]
F[k]     = F_e + F_dev[k]

# outer lateral (z)  $\rightarrow$  theta_cmd
e_z[k]   = z_r[k] - z[k]
I_z[k]   = I_z[k-1] + kI_z*T_s*e_z[k]
dhat_z[k]= (dhat_z[k-1] + (e_z[k]-e_z[k-1])/T_s)/(1+sigma/T_s)
theta_cmd[k] = kP_z*e_z[k] + I_z[k] + kD_z*dhat_z[k]

# inner attitude (theta)
e_th[k]  = theta_cmd[k] - theta[k]
dhat_th  = (dhat_th[k-1] + (e_th[k]-e_th[k-1])/T_s)/(1+sigma/T_s)
tau[k]   = kP_th*e_th[k] + kD_th*dhat_th
```

Implementation: `hw1_s/F_planar_vtol/ctrlNestedPID_digital.py:1` (includes hover force F_e and mixing to motor forces).

Checklist:

- Step in h and z : $SSE \rightarrow 0$.
- Inner θ loop faster than outer loops.
- Works under $\pm 20\%$ variation in (m_c, J_c, d, μ) when integrators are active.